

Developing phonetic typology through large-scale speech corpora

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Recent growth in the availability of large-scale speech corpora, computational tools for phonetic analysis, and processing capacity has created new opportunities to investigate crosslinguistic phonetic patterns at an unprecedented scale. In this talk, I introduce the VoxCommunis Corpus, a multilingual speech resource derived from Mozilla Common Voice that currently includes analysis-ready data from more than 70 languages. I then present two studies illustrating how large-scale corpus analyses can inform our understanding of phonetic typology and the mechanisms underlying crosslinguistic regularities.

The first case study revisits vowel dispersion, a foundational topic in phonetic and phonological typology (e.g., Liljencrants & Lindblom, 1972; Lindblom, 1986; Schwartz et al., 1997). Theories of adaptive dispersion predict that languages with larger vowel inventories should realize their vowel categories with greater precision and, potentially, greater peripheralization. Across a broad sample of languages, we find that larger vowel inventories are associated with more precise phonetic realization, but not with greater peripheralization. The results provide support for dispersion while also demonstrating how conclusions based on smaller language samples may be sensitive to sampling variability (Zhang & Chodroff, in prep).

The second study investigates intrinsic vowel duration, a proposed mechanistic phonetic universal whereby high vowels tend to be shorter than low vowels across languages. This pattern is commonly attributed to biomechanical factors: low vowels require greater jaw displacement, which may increase the time needed to reach the articulatory target (Solé & Ohala, 2010). Our analysis provides broad empirical support for the crosslinguistic prevalence of the duration difference (Chodroff & Zhang, in press). However, we argue that biomechanics alone may not fully account for its consistency. The explanation may additionally require a uniformity constraint on phonetic realization, i.e., the representations that serve as input to speech-motor commands, such that phonetic targets are held relatively constant across vowel categories. This constraint could in turn interact with language-specific functional pressures, such as contrast enhancement, although we find only limited evidence for such enhancement in the present data.

Together, these studies demonstrate both the potential and the challenges of using multilingual speech corpora to identify crosslinguistic phonetic regularities, reassess established typological claims, and evaluate the explanatory mechanisms proposed to underlie them.

Chodroff, E. & Zhang, M. (in press). A crosslinguistic corpus phonetic analysis of intrinsic vowel duration. In press at *Journal of Phonetics*.

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Solé, M. J., & Ohala, J. J. (2010). What is and what is not under the control of the speaker: Intrinsic vowel duration. *Papers in Laboratory Phonology*, 10, 607-655.

Zhang, M. & Chodroff, E. (in prep). Vowel dispersion revisited: Evidence from 67 languages.